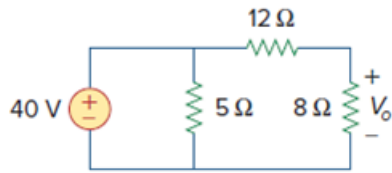


## Problem

Assume that  $V_o = 1$  V and use linearity to calculate the actual value of  $V_o$  in the circuit of Fig. 4.5.

Figure. 4.5:



## Step-by-step solution

## Step 1 of 4

Draw the circuit diagram shown in Figure 1.

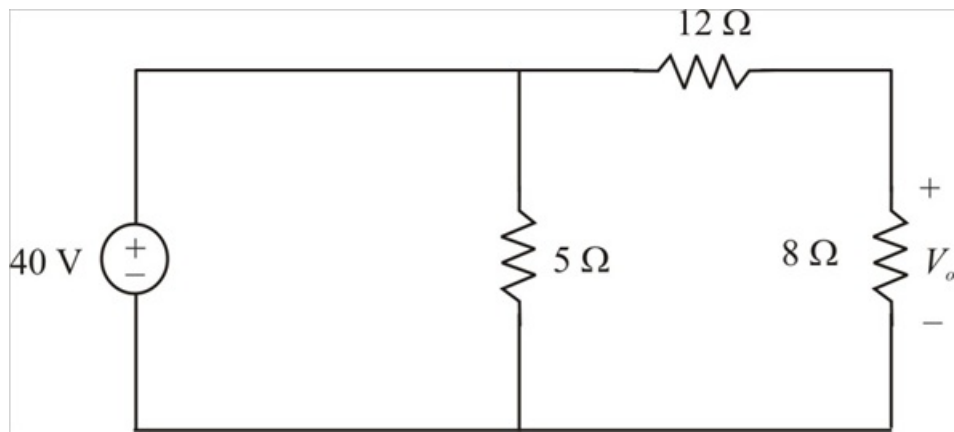


Figure 1

## Step 2 of 4

The circuit is labeled as shown in Figure 2.

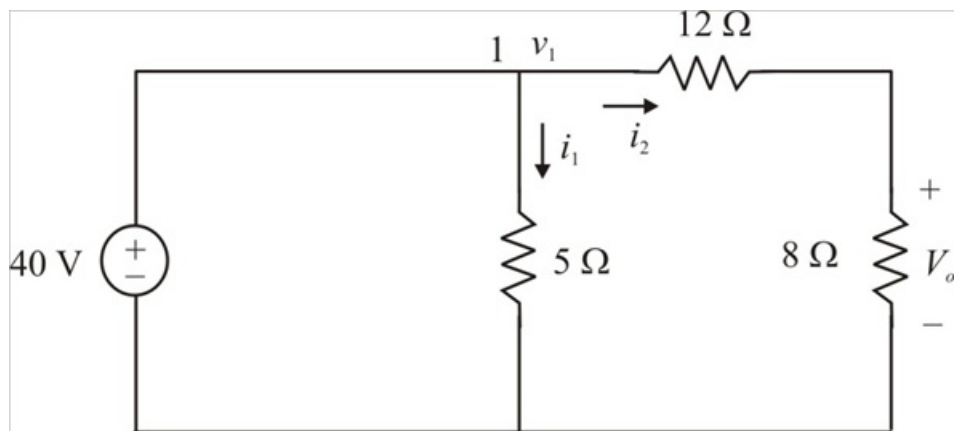


Figure 2

## Step 3 of 4

If  $V_o = 1\text{ V}$ , then the current flowing through  $8\text{ k}\Omega$  resistor is

$$\begin{aligned} i_{8\Omega} &= \frac{V_o}{R} \\ &= \frac{1}{8} \\ &= 125\text{ mA} \end{aligned}$$

The voltage drop across  $12\text{ k}\Omega$  resistor is

$$\begin{aligned} v_{12\Omega} &= i_2 R \\ &= (125 \times 10^{-3})(12) \\ &= 1.5\text{ V} \end{aligned}$$

#### Step 4 of 4

Thus the voltage at node 1 is

$$\begin{aligned} v_1 &= V_o + v_{12\Omega} \\ &= 1 + 1.5 \\ &= 2.5\text{ V} \end{aligned}$$

But,

$$\begin{aligned} V_s &= v_1 \\ &= 2.5\text{ V} \end{aligned}$$

This shows assuming  $V_o = 1\text{ V}$ , gives  $V_s = 2.5\text{ V}$ . The actual source voltage of  $40\text{ V}$  will give  $V_o$  as,

$$\begin{aligned} V_o &= \frac{40}{2.5} \\ &= 16\text{ V} \end{aligned}$$

Therefore the voltage  $V_o$  is  $16\text{ V}$ .